

Large cardinals at the inconsistency frontier: thin sections and saturation

A cardinal-sized strengthening of the finite-transversal obstruction, with detailed proofs and an I0-equiconsistency refinement

Prepared for Vladimir Reshetnikov

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Abstract. Let D_λ be the cofinal subsets of λ of order type ω , and let $Q_\lambda = D_\lambda / =^*$, where $=^*$ is equality modulo finite difference. We strengthen the finite-valued transversal obstruction to arbitrary fibers of cardinality below λ , with a common family of classes for all low-rank parameters. If λ is ultraexacting, then at least λ classes $q \in Q_\lambda$ satisfy

$$T \in \text{OD}_{V_\lambda}, \quad T \subseteq D_\lambda \quad \implies \quad |T \cap q| \in \{0, \lambda\}.$$

The same classes work for every such T , even with different parameters from V_λ and arbitrarily large ordinal parameters. The proof combines fixed small cofinal unions, a fixed bijection $\lambda \rightarrow V_\lambda$, and uniform bounded-rank correctness of ordinal definability. Both the fiber-cardinality cutoff and the uniform parameter-rank cutoff are sharp. We also exclude nonempty OD_{V_λ} families of fewer than λ thin complete sections, without requiring their members definable or uniformly bounding their fibers. Further results give a preserved-predicate obstruction to an Icarus embedding, bounded-rank separation under ordinary exactingness, and an I0-equiconsistent model with this saturation, $V_\lambda \subseteq \text{HOD}$, and definable choice on the bounded quotient.

What is established, and what is not

The main conclusions are **conditional inconsistency theorems in ZFC**: ultraexactingness is incompatible with the stated definable thinness principles. Together with results of Aguilera–Bagaria–Goldberg–Lücke, they give an I0-equiconsistent model with the new saturation properties and $V_\lambda \subseteq \text{HOD}$.

The new statements strengthen the supplied reports; worldwide priority has not been established. The proofs are conventional and have not been independently refereed or machine-checked. No unconditional inconsistency of a standard large-cardinal hypothesis is claimed.

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1 The result and its relationship to the supplied research

The two supplied documents play different roles. The *Unified Report* [6] maps several consistency boundaries. The *Synthesis* [5] consolidates nine continuations, including cofinality barriers for cover exactingness, finite-choice obstructions at ultraexacting cardinals, and consistency calibrations. Repeating those results under new names would not advance the investigation. This continuation concentrates on the finite-valued transversal theorem in its Section 9 and asks how much of its finiteness hypothesis is genuinely needed.

The essential observation is elementary but stronger than finite periodicity: *the union of fewer than λ countable sets has cardinality below λ* . If such a family of cofinal sets is preserved by an exacting embedding, its union is a preserved short cofinal subset of λ , contradicting the critical-sequence obstruction. This works even though $\text{cf}(\lambda) = \omega$; no regularity of λ in V is being assumed.

The harder part is the quantifier order. A particular ordinal-definable set need not be fixed by a chosen embedding, since the ordinal parameters in its definition need not be fixed. We handle all definitions simultaneously by stabilizing bounded-rank ordinal definability before choosing the embedding, and then selecting a canonical least counterexample from the fixed parameters. Section 4 supplies the full bookkeeping.

1.1 Objects and terminology

For an uncountable cardinal λ with $\text{cf}(\lambda) = \omega$, set

$$D_\lambda = \{a \subseteq \lambda : \text{ot}(a) = \omega \text{ and } \sup a = \lambda\}, \quad (1.1)$$

$$a =^* b \iff |a \Delta b| < \omega, \quad (1.2)$$

$$[a] = \{b \in D_\lambda : b =^* a\}, \quad Q_\lambda = \{[a] : a \in D_\lambda\}. \quad (1.3)$$

Here OD_p means definability from the *single set parameter* p and finitely many ordinals. By contrast, OD_{V_λ} allows finitely many parameters that are elements of V_λ . A finite tuple of such parameters can be coded by one element of V_λ .

Definition 1.1. A *complete section* is a set $T \subseteq D_\lambda$ meeting every class in Q_λ . It is *thin* if

$$0 < |T \cap q| < \lambda \quad (q \in Q_\lambda).$$

There is no requirement that one cardinal below λ bound all these intersections. A transversal is a complete section meeting each class exactly once.

Definition 1.2. For any set parameter p , define the set of *fully saturated classes*

$$\mathcal{G}_p(\lambda) = \{q \in Q_\lambda : \forall T \in \text{OD}_p (T \subseteq D_\lambda \implies |T \cap q| \in \{0, \lambda\})\}.$$

“Saturated” in this report refers only to this explicitly defined property, not to saturation of an ideal, a model, or a Boolean algebra.

Definition 1.3 (Simultaneity over all low-rank parameters). Let

$$\mathcal{G}_{\text{low}}(\lambda) = \{q \in Q_\lambda : \forall T \in \text{OD}_{V_\lambda} (T \subseteq D_\lambda \implies |T \cap q| \in \{0, \lambda\})\}.$$

Thus $\mathcal{G}_{\text{low}}(\lambda) = \bigcap_{p \in V_\lambda} \mathcal{G}_p(\lambda)$.

Theorem 1.4 (Main saturation theorem). In ZFC, if λ is ultraexacting, then

$$|\mathcal{G}_{\text{low}}(\lambda)| \geq \lambda.$$

In particular, no thin complete section belongs to OD_{V_λ} .

The theorem will be proved as Theorem 5.4. Its first conclusion supplies a common set of classes for *all* definitions from elements of V_λ , not just all definitions with one fixed parameter. The proof constructs the common classes directly: it does not infer a large intersection merely from knowing that each $\mathcal{G}_p(\lambda)$ is large.

1.2 An explicit conditional inconsistency statement

Write $\text{ThinCS}(\lambda)$ for

$$\exists p \in V_\lambda \exists T \in \text{OD}_p \quad (T \subseteq D_\lambda \wedge \forall q \in Q_\lambda (0 < |T \cap q| < \lambda)).$$

Then the central negative result is the ordinary first-order implication

$$\text{ZFC} \vdash \neg \exists \lambda (\text{UEx}(\lambda) \wedge \text{ThinCS}(\lambda)). \quad (1.4)$$

Ordinal definability is first-order definable, so this is not an unspecified scheme with a new truth predicate. An even stronger refutation, involving small families of potentially nondefinable sections, appears in Section 7.

2 Framework and external inputs

We work in an ambient universe satisfying ZFC. Cardinalities, power sets, and cofinality are ambient unless an inner model is explicitly indicated. All witnesses in the main argument are *set* embeddings between set structures; their elementarity is expressed using the usual satisfaction relation for sets. Their domains need not be transitive.

Definition 2.1 (Exacting and ultraexacting). A cardinal λ is exacting if, at every sufficiently large rank height ζ , there are

$$X < V_\zeta, \quad V_\lambda \cup \{\lambda\} \subseteq X, \quad j : X \longrightarrow V_\zeta$$

with j elementary, $j(\lambda) = \lambda$, and $j \restriction \lambda \neq \text{id}$. It is ultraexacting if these witnesses can also satisfy $j \restriction V_\lambda \in X$. We use the equivalent all-height/high-critical-point formulation in Input B.

The primary literature provides the following inputs. They are separated from the new proofs so that the external dependencies are visible.

Input A (Local Kunen inconsistency). There is no nontrivial elementary self-embedding of $V_{\mu+2}$ in ZFC. [3, 4]

Input B (Choice of witnesses). If λ is ultraexacting, then for any sufficiently large height ζ and any $\beta < \lambda$, an ultraexacting witness can be chosen with $\text{crit}(j) > \beta$. The corresponding high-critical-point formulation also holds for ordinary exacting witnesses. [1, Lemmas 2.3 and 3.2]

The strict inequality is obtained from the cited identity-on-an-initial-segment form by applying it to $\beta + 1$. In particular, choosing the critical point above the rank of $p \in V_\lambda$ makes $j(p) = p$.

Input C (Consistency comparison and coding). The theories $\text{ZFC} + \exists \lambda \text{ UEx}(\lambda)$ and $\text{ZFC} + \text{I0}$ are equiconsistent. Moreover, an ultraextending λ can be preserved by a set forcing that makes $V_\lambda \subseteq \text{HOD}$. [2, Theorem 4.5 and Proposition 3.9]

These sources were consulted directly for the stated theorem and lemma numbers. The argument below does not use the synthesis's $\text{I3}_{\text{wf}}(0)$ forcing construction, its high-completeness core theorem, or its unbounded-range obstruction for class embeddings. Those separate lines are not dependencies of the results below.

2.1 What “detailed proofs” covers here

Every new implication is proved below, including the canonical-definition argument, the construction of many fixed equivalence classes, the singular cardinal estimate for nonuniform families, and both directions of the consistency refinement. Input C is an imported large-cardinal comparison, not reproved or presented as a new comparison. Ordinary background facts about set satisfaction, rank, elementary embeddings, and finite coding are used explicitly where needed.

3 The local obstruction: fixed small cofinal unions

Lemma 3.1 (Critical-sequence normalization). Let $j : X \rightarrow V_\zeta$ be an exacting witness at λ , where $\zeta > \lambda + \omega$. Then $e = j \upharpoonright V_\lambda$ is a nontrivial elementary self-embedding of V_λ . If

$$\kappa = \text{crit}(e), \quad \kappa_n = e^n(\kappa) \quad (n < \omega),$$

then

$$\sup_{n < \omega} \kappa_n = \lambda, \quad e(\xi) > \xi \quad (\kappa \leq \xi < \lambda).$$

Also j fixes every element of V_κ .

Proof. The rank V_λ is definable from λ . Since $\lambda \in X$, $V_\lambda \in X$; since $V_\lambda \subseteq X$ as well, satisfaction in that rank is computed over the full structure, not a truncated domain. Relativizing any formula to V_λ and using $j(\lambda) = \lambda$ therefore proves the elementarity of e .

Its critical point is a cardinal, and is measurable by the usual derived ultrafilter

$$U = \{A \subseteq \kappa : \kappa \in e(A)\}.$$

All subsets and the short sequences needed to verify κ -completeness have rank below λ . Thus κ is strongly inaccessible. Elementarity gives that each κ_n is strongly inaccessible as well. In particular the sequence is strictly increasing and bounded by λ .

Suppose $\mu = \sup_n \kappa_n < \lambda$. The whole sequence then has rank below λ ; applying e shifts it by one place, so $e(\mu) = \mu$. Since λ is a cardinal and $\mu < \lambda$, the rank $V_{\mu+2}$ is still below λ . The restriction of e to this definable, fixed rank is a nontrivial elementary self-embedding, contrary to Input A. Hence the supremum is λ .

For $\kappa \leq \xi < \lambda$, choose n with $\kappa_n \leq \xi < \kappa_{n+1}$. Then $e(\xi) \geq e(\kappa_n) = \kappa_{n+1} > \xi$. Finally, the usual rank induction below the inaccessible κ proves that $e \upharpoonright V_\kappa = \text{id}$: each set of rank below κ has size below κ , so its image is its pointwise image; apply the induction hypothesis to its elements. This also gives the last assertion for j . $\square$

In particular, $\text{cf}^V(\lambda) = \omega$ and λ is a strong limit, as a limit of inaccessible cardinals. The only fixed ordinals below λ are those below κ .

Lemma 3.2 (Fixed small sets of ordinals). In the setting of Lemma 3.1, suppose $A \in X$, $A \subseteq \lambda$, $j(A) = A$, and $|A| < \lambda$. Then $A \subseteq \kappa$. Consequently, no fixed cofinal subset of λ has cardinality below λ .

Proof. Put $\tau = \text{ot}(A)$. Since A is a subset of the cardinal λ , $\tau \leq \lambda$; the size assumption excludes equality, so $\tau < \lambda$. The order type and the unique increasing enumeration $h : \tau \rightarrow A$ are definable from A , and therefore belong to X . Because A is fixed, $j(\tau) = \tau$ and $j(h) = h$. Lemma 3.1 gives $\tau < \kappa$. For every $\xi < \tau$, all the following evaluations are defined:

$$j(h(\xi)) = j(h)(j(\xi)) = h(\xi).$$

Here $\xi \in X$ because $\xi < \lambda$, and $h(\xi) \in X$ by elementary closure under function evaluation. Thus every member of A is a fixed ordinal below λ , and hence is below κ . $\square$

Lemma 3.3 (Fixed small families of cofinal sets). In the same setting, there is no $B \in X$ satisfying

$$j(B) = B, \quad \emptyset \neq B \subseteq D_\lambda, \quad |B| < \lambda.$$

Proof. The set $A = \bigcup B$ belongs to X and is fixed by j , since union is a uniquely definable operation. It is cofinal in λ , because any member of B is cofinal. In the ambient universe, Choice gives

$$|A| \leq |B| \cdot \aleph_0 \leq \max\{|B|, \aleph_0\} < \lambda.$$

This contradicts Lemma 3.2. $\square$

Remark 3.4 (Why singularity causes no problem here). We are taking a union of a fixed number $\mu < \lambda$ of *countable* sets, not a union of μ sets whose cardinalities approach λ . The estimate $\mu \cdot \aleph_0 < \lambda$ is valid for every uncountable cardinal λ . Section 7 treats the genuinely nonuniform case separately rather than silently assuming regularity.

4 Ordinal parameters and uniform bounded-rank correctness

We need one precise device for turning a possibly non-fixed definable object into a fixed least counterexample. It is useful to make the device uniform before the critical-sequence class is known.

4.1 A canonical order on OD_p

Fix a set p . A local definition code consists of an ordinal θ with $p \in V_\theta$, a formula number, and a finite tuple of ordinals below θ . The code defines x when x is the unique element of V_θ satisfying that formula over $(V_\theta, \in)$ with those parameters and p . This is a first-order relation in the ambient universe, since the satisfaction relation is for a *set* structure.

Order these codes first by θ , and, at a fixed θ , by a fixed wellordering of formula numbers, finite lengths, and ordinal tuples. Each stage is a set. Every $x \in \text{OD}_p$ has such a code: apply reflection to a global definition of x and its uniqueness. Conversely, every locally coded set is in OD_p , since θ is an allowed ordinal parameter. Ordering sets by their least codes gives a definable, set-like class wellorder $<_p$ of OD_p .

Lemma 4.1 (Uniform bounded-rank stabilization). Fix a set P of possible parameters and an ordinal δ . There is a height η such that, at every sufficiently large limit cardinal $\zeta > \eta$, simultaneously for every $p \in P$, V_ζ computes correctly both

$$\text{OD}_p \cap V_\delta \quad \text{and} \quad <_p \restriction (\text{OD}_p \cap V_\delta).$$

In particular, one can take $P = \{p\}$ for a single prescribed parameter, or $P = V_{\lambda+1}$ before the embedding is chosen.

Proof. The code relation defines OD_p uniformly in p . Thus

$$S = \{(p, x) \in P \times V_\delta : x \in \text{OD}_p\}$$

is a set by Separation. To each $(p, x) \in S$ assign the unique least code of x relative to p . Replacement gives a set of all these codes and a bound on their heights and ranks. Choose η above this bound, above the ranks of P and V_δ , and above the finite coding overheads.

For a large enough limit cardinal $\zeta > \eta$, all the least codes under consideration are present in V_ζ , as are the rank structures and set satisfaction relations used in those codes. Evaluation of a local code is absolute. Hence every actual OD_p member of V_δ is recognized, uniformly for $p \in P$. Conversely, a code used by V_ζ is a genuine code in V : its structure is the full V_θ , and finite-formula satisfaction is absolute. There are no false positives. All codes preceding a least code have no larger height, so no earlier competing code is omitted. The least codes and their order are therefore correct, simultaneously for all the indicated parameters. $\square$

The lemma is not a truth predicate for V . It bounds a *set* of local codes for a previously fixed rank and a *set* of possible parameters. Its height can be arbitrarily large; ultraexactingness supplies witnesses at the height chosen afterwards. The uniformity in p will be essential: the parameter coding V_λ depends on the subsequently chosen embedding.

Lemma 4.2 (Fixed canonical counterexamples). Suppose $p, \lambda, q \in X$, $j(p) = p$, $j(\lambda) = \lambda$, and $j(q) = q$. Let $\Phi(T, p, \lambda, q)$ describe objects T in a fixed bounded rank, and assume V_ζ is correct about Φ , OD_p , and $<_p$ on those objects. If some $T \in \text{OD}_p$ satisfies Φ , then the $<_p$ -least such T belongs to X and is fixed by j .

Proof. Correctness makes the actual least object the unique solution of the same formula in V_ζ . By $X < V_\zeta$, that unique solution belongs to X . Apply j to its defining property. All displayed parameters are fixed, so $j(T)$ is the same unique solution in V_ζ . Hence $j(T) = T$. $\square$

For our applications, Φ is built from D_λ , Q_λ , finite symmetric difference, and cardinality below λ . At a height above $\lambda + \omega$, all relevant sets, increasing enumerations, and possible bijections with ordinals below λ are present. These properties are therefore absolute. Lemma 4.1, with $\delta = \lambda + 12$ and a suitable set P , supplies the remaining correctness uniformly for all candidate classes q and all allowed coding parameters that the subsequently chosen embedding might produce.

5 Many fixed classes and simultaneous saturation

Lemma 5.1 (The size of a finite-modification class). Every $q \in Q_\lambda$ has cardinality exactly λ .

Proof. For $q = [a]$, every member is determined by removing finitely many elements of a and adjoining finitely many ordinals below λ . There are at most λ such choices. Conversely, the sets $a \cup \{\xi\}$, for $\xi \in \lambda \setminus a$, are distinct members of $[a]$. Adjoining finitely many ordinals to an increasing cofinal ω -set leaves its order type ω : below any fixed bound only finitely many elements occur. There are λ choices of ξ , so the lower bound follows. $\square$

Lemma 5.2 (Internal shifted critical sequences). Let $j : X \rightarrow V_\zeta$ be an ultraexacting witness as above, and put $\kappa = \text{crit}(j)$ and $\kappa_n = j^n(\kappa)$. For every $t < \kappa$, the set

$$a_t = \{\kappa_n + t : n < \omega\}$$

and its class $q_t = [a_t]$ belong to X . Moreover,

$$\begin{aligned} j(a_t) &= a_t \setminus \{\kappa + t\}, & j(q_t) &= q_t, \\ a_t \cap a_s &= \emptyset & (t \neq s < \kappa). \end{aligned}$$

Proof. The restriction $e = j \upharpoonright V_\lambda$ is an element of X . Its critical point and its finite iterates on that critical point are uniformly definable from e . The recursion of length ω is available in V_ζ , and its unique resulting sequence therefore belongs to X . It is the actual critical sequence: all its values and finite evaluations occur in the full rank $V_\lambda \subseteq X$.

Since $t < \kappa$, it belongs to X and is fixed. Consequently a_t and its increasing enumeration belong to X . For a countable set with an enumeration in X , elementarity and the fact that j fixes every natural number imply that its image is its pointwise image. Using preservation of ordinal addition,

$$j(a_t) = \{j(\kappa_n + t) : n < \omega\} = \{\kappa_{n+1} + t : n < \omega\}.$$

This is the stated tail. Since the equivalence class is definable from a_t and the fixed λ , it is in X , and $j(q_t) = [j(a_t)] = [a_t]$.

Finally, the κ_n are increasing infinite cardinals. For $t < \kappa$, $\kappa_n + t < \kappa_n^+ \leq \kappa_{n+1}$. Thus the different n lie in disjoint intervals, and addition on the right is strictly increasing in t within an interval. The sets a_t are pairwise disjoint and cofinal, so their classes are pairwise distinct. $\square$

Lemma 5.3 (An internally available fixed code for the whole lower rank). For every ultraexacting witness as in Lemma 5.2, there is a bijection

$$b : \lambda \longrightarrow V_\lambda$$

such that $b \in X \cap V_{\lambda+1}$ and $j(b) = b$. Consequently,

$$\text{OD}_{V_\lambda} \subseteq \text{OD}_b.$$

This fixed-code construction is the same underlying device as the preserved wellordering used in [2, Proposition 3.9]; its proof is given here.

Proof. Let $\kappa = \text{crit}(j)$. Since κ is inaccessible, $|V_\kappa| = \kappa$. Choose a bijection $b_0 : \kappa \rightarrow V_\kappa$. Its graph has rank below λ , so $b_0 \in V_\lambda \subseteq X$. Put $e = j \upharpoonright V_\lambda$ and $b_n = e^n(b_0)$. Each b_n is a bijection $\kappa_n \rightarrow V_{\kappa_n}$.

For $\xi < \kappa$, both ξ and $b_0(\xi)$ are fixed by j , so

$$b_1(\xi) = j(b_0)(\xi) = j(b_0(\xi)) = b_0(\xi).$$

Thus $b_0 \subseteq b_1$. Applying e successively shows $b_n \subseteq b_{n+1}$ for every n . Since $e, b_0 \in X$, the sequence $\langle b_n : n < \omega \rangle$ is uniquely definable by recursion and belongs to X . Its union b belongs to X . The domains increase to λ and the ranges increase to V_λ , so b is a bijection between them.

The countable sequence is in X , its indices are fixed, and $j(b_n) = b_{n+1}$. Therefore

$$j(b) = \bigcup_{n < \omega} b_{n+1} = \bigcup_{n < \omega} b_n = b.$$

Each ordered pair in the graph of b has rank below the limit cardinal λ , whence $b \subseteq V_\lambda$ and $b \in V_{\lambda+1}$.

Finally, every $p \in V_\lambda$ equals $b(\xi)$ for some ordinal $\xi < \lambda$. A definition using finitely many such parameters can replace each one by its uniquely specified value under b . The result is a definition from b and finitely many ordinals. This proves the inclusion. Notice that the coding parameter b need not itself belong to V_λ ; its fixedness is proved directly, not inferred from its rank. $\square$

Theorem 5.4 (Simultaneous full-sized traces for all low-rank parameters). If λ is ultraexacting, then

$$|\mathcal{G}_{\text{low}}(\lambda)| \geq \lambda.$$

Thus at least λ classes have only empty or full-sized intersections with every OD_{V_λ} subset of D_λ .

Proof. Fix an arbitrary $\beta < \lambda$. Apply Lemma 4.1 with

$$P = V_{\lambda+1}, \quad \delta = \lambda + 12,$$

and choose a sufficiently large limit cardinal ζ above its bound. Use Input B to obtain an ultraexacting witness $j : X \rightarrow V_\zeta$ with $\kappa = \text{crit}(j) > \beta$. Lemma 5.3 supplies a fixed bijection $b \in X \cap V_{\lambda+1}$ coding all of V_λ . For each $t < \kappa$, Lemma 5.2 supplies a fixed class $q_t \in X$.

We claim that $q_t \in \mathcal{G}_b(\lambda)$ for each $t < \kappa$. If not, the fact that $|q_t| = \lambda$ gives some $T \in \text{OD}_b$ satisfying

$$T \subseteq D_\lambda, \quad 0 < |T \cap q_t| < \lambda. \quad (5.1)$$

Although b was unknown when ζ was chosen, it belongs to the set P of parameters for which correctness was made uniform. Thus V_ζ is correct about OD_b , its canonical order, and (5.1). Since b, λ, q_t are fixed, the canonical least witness T_* is a fixed member of X , by Lemma 4.2. Then

$$B = T_* \cap q_t$$

is a fixed member of X , is a nonempty family of members of D_λ , and has cardinality below λ . This contradicts Lemma 3.3.

Consequently all the q_t are in $\mathcal{G}_b(\lambda)$. The inclusion $\text{OD}_{V_\lambda} \subseteq \text{OD}_b$ implies

$$\mathcal{G}_b(\lambda) \subseteq \mathcal{G}_{\text{low}}(\lambda).$$

The κ classes q_t are pairwise distinct, so $|\mathcal{G}_{\text{low}}(\lambda)| \geq \kappa > \beta$. Since $\beta < \lambda$ was arbitrary, the desired bound follows. Explicitly, a hypothetical cardinality $\mu < \lambda$ would be contradicted by the same construction with $\beta = \mu$. $\square$

Remark 5.5 (How the quantifiers are handled). No intersection principle is being assumed. A single fixed coding parameter b covers *every* low-rank parameter. Correctness is uniform over all possible b before the embedding is chosen. The least-counterexample argument then works for arbitrary ordinal parameters, even ones that j moves. These three ingredients give the common set of classes directly.

Corollary 5.6 (Fixed parameters and bounded ranks). For each $p \in V_\lambda$, $|\mathcal{G}_p(\lambda)| \geq \lambda$. For every $\alpha < \lambda$, at least λ classes also have only empty or λ -sized intersections with all OD_{V_α} subsets of D_λ . In both assertions the same set $\mathcal{G}_{\text{low}}(\lambda)$ works simultaneously.

Proof. Both OD_p and OD_{V_α} are subclasses of OD_{V_λ} . Apply Theorem 5.4 and the definition of $\mathcal{G}_{\text{low}}(\lambda)$. $\square$

6 Thinness principles that are refuted

Corollary 6.1 (No definable thin complete section). At an ultraexacting λ , no $T \in \text{OD}_{V_\lambda}$ is a thin complete section of Q_λ . In particular this excludes transversals, finite-valued complete sections, countable-valued complete sections, and sections whose fiber sizes vary below λ without a common bound.

Proof. Pick $q \in \mathcal{G}_{\text{low}}(\lambda)$. Completeness makes $T \cap q$ nonempty, and goodness makes its cardinality λ , contradicting thinness. $\square$

Corollary 6.2 (A common saturation set). Every OD_{V_λ} complete section T satisfies

$$\forall q \in \mathcal{G}_{\text{low}}(\lambda) \quad |T \cap q| = \lambda.$$

The same set $\mathcal{G}_{\text{low}}(\lambda)$ works for all such T and has cardinality at least λ .

Proof. This is the definition of goodness together with completeness and Theorem 5.4. $\square$

Corollary 6.3 (No local definable thin coverage). The following apparently weaker requirement fails:

$$\forall q \in Q_\lambda \exists T \in \text{OD}_{V_\lambda} \quad (T \subseteq D_\lambda \wedge 0 < |T \cap q| < \lambda).$$

The set T is allowed to depend on q and need not be a complete section.

Proof. Every member of $\mathcal{G}_{\text{low}}(\lambda)$ witnesses failure. $\square$

Corollary 6.4 (Ordinal-valued maps have large fibers). If $f : D_\lambda \rightarrow \text{Ord}$ is a set function in OD_{V_λ} , then for every $q \in \mathcal{G}_{\text{low}}(\lambda)$ and every ordinal ξ ,

$$|q \cap f^{-1}\{\xi\}| \in \{0, \lambda\}.$$

Thus there is no OD_{V_λ} ordinal-valued map whose nonempty fibers inside every equivalence class all have cardinality below λ .

Proof. The set $T_\xi = f^{-1}\{\xi\}$ belongs to OD_{V_λ} : the additional parameter ξ is ordinal and is allowed. Apply the definition of $\mathcal{G}_{\text{low}}(\lambda)$. For the last claim, any q has at least one value of f , so at least one of these fibers is nonempty. $\square$

6.1 Sharp cardinal and parameter cutoffs

Proposition 6.5 (The cardinal cutoff is sharp). The bound $< \lambda$ in Corollary 6.1 cannot be replaced by $\leq \lambda$: $T = D_\lambda$ is an ordinal-definable complete section and $|T \cap q| = \lambda$ for every $q \in Q_\lambda$.

Proof. The set D_λ is definable from the ordinal λ , and the size assertion is Lemma 5.1. $\square$

Proposition 6.6 (The uniform parameter-rank cutoff is sharp). No class $q \in Q_\lambda$ has only empty or λ -sized intersections with all $\text{OD}_{V_{\lambda+1}}$ subsets of D_λ . Equivalently,

$$\bigcap_{p \in V_{\lambda+1}} \mathcal{G}_p(\lambda) = \emptyset.$$

Thus V_λ in the simultaneous theorem cannot be replaced by $V_{\lambda+1}$.

Proof. For any $q \in Q_\lambda$, choose $a \in q$. The subset $a \subseteq \lambda$ belongs to $V_{\lambda+1}$. Taking the allowed parameter $p = a$, the set $T = \{a\}$ belongs to OD_p and satisfies $|T \cap q| = 1 < \lambda$. Hence $q \notin \mathcal{G}_p(\lambda)$. $\square$

7 Small families without a uniform fiber bound

The preceding results concern a single definable set of representatives. We now allow the sections themselves to be nondefinable. Only the family containing them is required to be ordinal-definable from low-rank parameters. The new difficulty is that a union of fewer than λ many sets of size below λ can have size λ , since $\text{cf}(\lambda) = \omega$. The embedding itself supplies the bound that ambient cardinal arithmetic alone does not supply.

Theorem 7.1 (Local obstruction to a small family of thin sections). Let $j : X \rightarrow V_\zeta$ be an ultraexacting witness at λ , with $\zeta > \lambda + \omega$. There is no fixed set $\mathcal{F} \in X$ such that

$$\emptyset \neq \mathcal{F}, \quad |\mathcal{F}| < \lambda, \quad \text{every } T \in \mathcal{F} \text{ is a thin complete section of } Q_\lambda.$$

No assumption is made that the members of $\mathcal{F}$ belong individually to X , are fixed, or have their fibers bounded by one common cardinal.

Proof. Let $\kappa = \text{crit}(j)$ and let $q = q_0$ be the fixed critical-sequence class from Lemma 5.2. Form the set of actual cardinal numbers

$$A = \{|T \cap q| : T \in \mathcal{F}\}.$$

Thinness gives $A \subseteq \lambda$, and $|A| \leq |\mathcal{F}| < \lambda$. All the sets and possible bijections needed to compute these cardinalities have rank below ζ . Consequently A is uniquely definable in V_ζ from $\mathcal{F}, q$, so $A \in X$. Since both these parameters are fixed, $j(A) = A$. Lemma 3.2 gives

$$A \subseteq \kappa. \tag{7.1}$$

Thus a uniform bound has been obtained *at this class*: every $T \in \mathcal{F}$ has $|T \cap q| < \kappa$.

Now put

$$B = q \cap \bigcup_{T \in \mathcal{F}} T = \bigcup_{T \in \mathcal{F}} (T \cap q).$$

This is another fixed member of X . It is nonempty, because $\mathcal{F}$ is nonempty and every one of its sections meets q . It is a family of cofinal ω -subsets of λ . In the ambient universe, writing $\mu = |\mathcal{F}| < \lambda$, Choice and (7.1) give

$$|B| \leq \mu \cdot \kappa < \lambda.$$

The last inequality uses only that $\mu, \kappa < \lambda$ and that κ is infinite: their product is at most $\max\{\mu, \kappa\}$ when μ is infinite, and at most κ otherwise. We have contradicted Lemma 3.3. $\square$

Theorem 7.2 (No definable small family of thin sections). If λ is ultraexacting, there is no nonempty family $\mathcal{F} \in \text{OD}_{V_\lambda}$ of cardinality below λ all of whose members are thin complete sections of Q_λ .

Proof. Suppose such a family exists, with $\mathcal{F} \in \text{OD}_p$ for some $p \in V_\lambda$. Every candidate family is a subset of $\mathcal{P}(D_\lambda)$, so all candidates lie in $V_{\lambda+12}$. Stabilize OD_p and its canonical order on this rank using Lemma 4.1 with $P = \{p\}$, and choose an ultraexacting witness at the resulting sufficiently large height, with critical point above $\text{rank}(p)$.

The property “nonempty, of size below λ , and consisting of thin complete sections” is correct at this height: it quantifies over the full D_λ and Q_λ , and its cardinality comparisons use bijections of bounded rank. The canonical least OD_p family having that property is therefore a fixed member of X , by the proof of Lemma 4.2 with parameters p, λ . This contradicts Theorem 7.1. $\square$

Corollary 7.3 (A stronger inconsistent theory). The following theory is inconsistent:

$$\begin{aligned} &\text{ZFC} + “\lambda \text{ is ultraexacting}” \\ &+ “\text{there is a nonempty } \text{OD}_{V_\lambda} \text{ family of fewer than } \lambda \\ &\quad \text{thin complete sections of } Q_\lambda”. \end{aligned}$$

In particular, a countable definable family of possibly nondefinable transversals is impossible.

Proof. The theorem excludes precisely the displayed conjunction. A transversal is thin, and a countable family has size below the uncountable λ . $\square$

Remark 7.4 (Which earlier question this does not settle). A thin complete section is a subset of the *representative space* D_λ . A selector on finite subsets of Q_λ chooses one *equivalence class* from each finite collection of classes. These are different types of objects. The synthesis’s question about a countable or $< \lambda$ -sized definable family of such selectors is not resolved by Theorem 7.2. No reduction from that problem to the present one has been proved.

8 A preserved thin predicate obstructs an Icarus embedding

The local argument does not require the whole exacting framework. If an inner model already contains every subset of V_λ , the critical sequence is available there without an ultraexacting domain hypothesis. This gives a precise obstruction to enriching a rank-into-rank inner model by a thin complete section that the embedding must preserve.

Theorem 8.1 (Thin-predicate obstruction). Work in an ambient universe satisfying ZFC. Let λ be an uncountable cardinal, and let N be a transitive inner model of ZF such that $V_{\lambda+1} \subseteq N$. Suppose there is an elementary embedding $j : N \rightarrow N$ with critical point $\kappa < \lambda$ whose critical sequence is a set and satisfies

$$\sup_{n < \omega} j^n(\kappa) = \lambda.$$

Then j cannot preserve any thin complete section $T \in N$ of Q_λ :

$$\text{Thin}(T, \lambda) \implies j(T) \neq T.$$

Thinness and its cardinal bounds are measured in the ambient universe. No form of Choice in N is assumed.

Proof. Write $\kappa_n = j^n(\kappa)$. The sequence $c = \langle \kappa_n : n < \omega \rangle$ is a subset of V_λ , hence belongs to $V_{\lambda+1}$ and to N . Its range $a = \{\kappa_n : n < \omega\}$ belongs to N as well. Elementarity and the fact that j fixes ω give

$$j(c)(n) = \kappa_{n+1}, \quad j(a) = a \setminus \{\kappa\}, \quad j(\lambda) = \lambda.$$

The last equality follows by applying j to $\lambda = \sup \text{ran}(c)$ inside N .

The model N computes D_λ correctly. Indeed, it contains all subsets of λ ; order type ω and cofinality in this definition are absolute for these wellordered subsets. Their increasing enumerations are uniquely definable and belong to N . Thus N also computes the full class $q = [a]$ correctly, and $q \in N$. Finite modification gives $j(q) = q$.

Suppose $T \in N$ is a thin complete section and $j(T) = T$. Then

$$B = T \cap q, \quad A = \bigcup B$$

are fixed sets in N . Ambient thinness makes B a nonempty family of fewer than λ countable sets. Ambient Choice therefore gives $|A| < \lambda$, while A is cofinal in λ .

Let $\tau = \text{ot}(A)$, computed in either transitive universe; this ordinal is absolute. Since λ is a cardinal and $|A| < \lambda$, we have $\tau < \lambda$. The increasing enumeration $h : \tau \rightarrow A$ belongs to N , and $j(A) = A$ implies $j(\tau) = \tau$ and $j(h) = h$. Every ordinal $\xi \in [\kappa, \lambda)$ is moved: if $\kappa_n \leq \xi < \kappa_{n+1}$, then $j(\xi) \geq \kappa_{n+1} > \xi$. Hence $\tau < \kappa$. For $\xi < \tau$,

$$j(h(\xi)) = j(h)(j(\xi)) = h(\xi),$$

so every element of A lies below κ . This contradicts its cofinality. $\square$

Corollary 8.2 (An explicit forbidden enrichment). Let T be any actual thin complete section of Q_λ . There is no elementary self-embedding of the structure

$$(L(V_{\lambda+1}, T), \in, T)$$

with critical point below λ and critical sequence cofinal in λ . Here the final T is a named set, so its preservation is part of the structure's elementarity requirement.

Proof. Take $N = L(V_{\lambda+1}, T)$, understood as the constructible inner model containing $V_{\lambda+1}$ and T as sets. Its transitivity gives $V_{\lambda+1} \subseteq N$, and naming T requires $j(T) = T$. Theorem 8.1 applies. $\square$

There are thin complete sections in the ambient universe by Choice: choose one representative from each member of the set Q_λ . The corollary therefore excludes genuine enrichments, not an empty collection of possible parameters. On the other hand, the preservation requirement is essential to the argument. An embedding of the underlying inner model that moves T is not refuted. Nor does the argument refute I0, whose model need not contain a thin section that is fixed by its embedding.

9 A bounded-rank separation theorem from ordinary exactingness

The saturation theorem used ultraexactingness to put the critical sequence inside the domain. A different consequence needs only exactingness. It clarifies why a small definable family of objects of rank $\lambda + 1$ cannot hide its distinctions at ranks cofinal in λ .

Lemma 9.1 (The ordinary exacting cofinality barrier). *If λ is exacting and $S \in \text{OD}_{V_\lambda}$ is a subset of λ with $|S| < \lambda$, then $\sup S < \lambda$.*

Proof. Suppose instead that such a set is cofinal, and choose $p \in V_\lambda$ such that some cofinal $S \in \text{OD}_p$ has size below λ . Stabilize the canonical OD_p order on a rank containing all subsets of λ , and choose an ordinary exacting witness with critical point above $\text{rank}(p)$. At this height the property “a cofinal subset of λ of cardinality below λ ” is correct. Its canonical least OD_p witness is fixed and belongs to X . Lemma 3.2 is valid for ordinary exacting witnesses and supplies the contradiction. $\square$

This is the familiar cofinality barrier, not a new large-cardinal comparison. Equivalently, λ is regular in HOD_{V_λ} , as proved in [1, Theorem 2.10]. For the direction needed here, a cofinal sequence in that inner model would have an OD_{V_λ} range of size below λ . We included the argument to make the separation proof independent of an unexplained appeal to inner-model regularity.

Theorem 9.2 (One bounded rank separates a small definable family). *Suppose λ is exacting,*

$$A \in \text{OD}_{V_\lambda}, \quad A \subseteq \mathcal{P}(V_\lambda) = V_{\lambda+1}, \quad |A| < \lambda.$$

Then some $\beta < \lambda$ makes the restriction map

$$r_\beta : A \longrightarrow \mathcal{P}(V_\beta), \quad r_\beta(x) = x \cap V_\beta$$

injective. Moreover,

$$A \in \text{HOD}_{V_\lambda}, \quad \text{and } A \text{ has an enumeration of length } |A| \text{ in } \text{HOD}_{V_\lambda}.$$

Here $|A|$ is its actual ambient cardinality, not a potentially larger inner-model cardinality.

Proof. For distinct $x, y \in A$, let

$$d(x, y) = \min\{\alpha < \lambda : x \cap V_\alpha \neq y \cap V_\alpha\}.$$

This minimum exists because $x, y \subseteq V_\lambda$: a member of their symmetric difference lies in some bounded rank. Put

$$S = \{d(x, y) : x, y \in A, x \neq y\}.$$

The definition of S uses A and the ordinal λ , so $S \in \text{OD}_{V_\lambda}$. Also $|S| \leq |A|^2 < \lambda$. Lemma 9.1 makes S bounded in λ . Choose $\beta < \lambda$ above all members of S ; when S is empty, any $\beta < \lambda$ suffices. If $x \neq y$, their restrictions already differ at $d(x, y)$ and still differ at β . This proves injectivity.

Write $H = \text{HOD}_{V_\lambda}$. For $x \in A$, its restriction $r = x \cap V_\beta$ belongs to $V_{\beta+1} \subseteq V_\lambda$. The object x is the unique member of A with this restriction. A definition of A from low-rank parameters, together with the new low-rank parameter r and the ordinal β , therefore defines x . Thus $x \in \text{OD}_{V_\lambda}$. Every element of x belongs to V_λ , and all of its transitive closure does too. These sets are hereditarily definable from allowed low-rank parameters. Hence $x \in H$. Since A itself is ordinal-definable from such parameters and all its members are hereditary in this sense, $A \in H$.

Choose in V a wellordering w of $\mathcal{P}(V_\beta)$. Its rank is below λ , so $w \in V_\lambda \subseteq H$. Pull it back along the injective map r_β to obtain a wellordering $\triangleleft$ of A in H . The order type τ is the actual order type, by transitivity, and has cardinality $|A| < \lambda$. Since λ is a cardinal, $\tau < \lambda$. The increasing enumeration $h : \tau \rightarrow A$ is computed correctly in H . Finally let $\mu = |A|$. An ambient bijection $b : \mu \rightarrow \tau$ has rank below λ , hence belongs to $V_\lambda \subseteq H$. The composition $h \circ b$ is an enumeration of A of length μ in H . The same argument handles the empty family with the empty enumeration. $\square$

The final proof does not assume that H satisfies global Choice. It constructs the required wellordering from a particular bounded-rank wellordering that is already an element of V_λ . The rank restriction on the members of A is also substantive: a selector defined on arbitrary equivalence classes has a graph of higher rank, and the above first-difference map need not take values below λ . This is another reason the unresolved selector problem does not follow from the separation theorem.

10 An I0-equiconsistent uniform-saturation package

The supplied reports already identify the consistency of ultraexactingness with a completely ordinal-definable rank below λ . The new theorem adds simultaneous low-parameter saturation to that model. Unlike the positive coding hypothesis, this saturation follows from ultraexactingness alone. Coding the lower rank into HOD then supplies a useful contrasting region where definable choice remains available.

Lemma 10.1 (Collapse of the parameter hierarchy). If $V_\lambda \subseteq \text{HOD}$, then

$$\text{OD}_{V_\lambda} = \text{OD}_p = \text{OD} \quad \text{for every } p \in V_\lambda.$$

If λ is also ultraexacting, then

$$\mathcal{G}_{\text{low}}(\lambda) = \mathcal{G}_p(\lambda) = \mathcal{G}_\emptyset(\lambda), \quad |\mathcal{G}_{\text{low}}(\lambda)| \geq \lambda.$$

Proof. An allowed parameter p is hereditarily ordinal-definable and therefore is itself ordinal-definable. Substitute a definition of p from its finite ordinal tuple into any definition using p and ordinal parameters. The result uses only finitely many ordinal parameters. The same substitution works for a finite list of elements of V_λ . This proves containment in OD; reverse containment is immediate by ignoring the extra parameters. The equalities of the three sets of classes follow from their definitions. The cardinal bound follows from Theorem 5.4. $\square$

10.1 A positive control below the cofinality boundary

Put

$$B_\lambda = \{a \subseteq \lambda : |a| = \aleph_0 \text{ and } \sup a < \lambda\}, \quad Q_\lambda^{\text{bd}} = B_\lambda / \equiv^*.$$

The members of B_λ are all bounded, in contrast to D_λ . They need not have order type ω : the displayed definition uses countable infinitude, which is sufficient for the control example.

Proposition 10.2 (The bounded quotient still has definable choice). If λ is an uncountable limit cardinal and $V_\lambda \subseteq \text{HOD}$, then Q_λ^{bd} has an ordinal-definable wellordering and an ordinal-definable transversal. These statements concern the full ambient bounded quotient, not merely a quotient computed using fewer representatives in HOD.

Proof. Every $a \in B_\lambda$ has rank below λ , so $a \in \text{HOD}$. Its countability has a witness of rank below λ : an enumeration of a bounded subset of λ by ω has bounded rank. Such a witness also belongs to HOD . Thus HOD recognizes exactly the same members of B_λ . Moreover B_λ is ordinal-definable from λ and all its elements are hereditary in HOD , so $B_\lambda \in \text{HOD}$.

For $a \in B_\lambda$, let $[a]_{\text{bd}}$ be its finite-modification class within B_λ . Every member of this class is bounded: finitely many additions to a bounded set remain bounded below the limit cardinal λ . The full class is definable from a, λ and consists of members of HOD . Since a is ordinal-definable, so is the class; it therefore belongs to HOD . The full quotient Q_λ^{bd} is ordinal-definable from λ and all its members belong to HOD , so it too belongs to HOD .

Restrict the canonical definable wellorder of HOD to the set Q_λ^{bd} . This gives an ordinal-definable wellordering. For each class, take its least representative in the canonical wellorder of HOD . Each class is nonempty and consists of HOD sets, so the least representative exists. The set of all these representatives is definable from the ordinal λ and meets each class exactly once. It is the required transversal. $\square$

This control separates two issues. The classes of the bounded quotient also have size λ , by the finite-addition argument of Lemma 5.1; their large size alone does not cause the obstruction. The obstruction uses the cofinality of *each representative* and the resulting cofinality of a fixed small union.

10.2 The formal comparison

Let T_∂ be the theory

$$\text{ZFC} + \exists \lambda (\text{UEx}(\lambda) \wedge V_\lambda \subseteq \text{HOD}).$$

Let T_{sat} assert that there is a cardinal λ with all the following properties, in addition to ZFC:

- (i) λ is ultraexacting and $V_\lambda \subseteq \text{HOD}$;
- (ii) $|\mathcal{G}_{\text{low}}(\lambda)| \geq \lambda$;
- (iii) there is no nonempty OD_{V_λ} family of fewer than λ thin complete sections of Q_λ ;
- (iv) the full bounded quotient Q_λ^{bd} has an ordinal-definable wellordering and an ordinal-definable transversal.

All these assertions are first-order expressible. In particular, local satisfaction codes express ordinal definability, and set embeddings express ultraexactingness. The usual I0 theory is understood in the formulation used in Input C.

Theorem 10.3 (Equiconsistency with simultaneous low-parameter saturation). The theories T_{sat} and T_∂ are equivalent over ZFC, and

$$\text{Con}(T_{\text{sat}}) \iff \text{Con}(\text{ZFC} + \text{I0}). \quad (10.1)$$

This is a refinement of the properties of a known consistency model, not a claim to lower or newly determine the consistency strength of ultraexactingness.

Proof. First, any model of T_{sat} satisfies T_∂ by clause (i). Conversely, let λ witness T_∂ . Theorem 5.4 gives clause (ii); Theorem 7.2 gives (iii); and Proposition 10.2 gives (iv). The same λ witnesses all clauses. Thus the two theories are equivalent.

For the forward consistency implication in (10.1), T_{sat} implies the existence of an ultraexacting cardinal. The ultraexacting-to-I0 direction of Input C then yields $\text{Con}(\text{ZFC} + \text{I0})$.

For the reverse implication, the other direction of Input **C** first gives the consistency of ZFC with an ultraexacting cardinal. Its preservation-and-coding clause supplies the relative consistency of an ultraexacting λ with $V_\lambda \subseteq \text{HOD}$, namely $\text{Con}(T_\partial)$. The proved implication $T_\partial \vdash T_{\text{sat}}$ yields $\text{Con}(T_{\text{sat}})$. These are the standard syntactic relative-consistency implications obtained from the cited comparison and forcing theorem; no transitive-model assumption is added to the hypothesis $\text{Con}(\text{ZFC} + \text{I0})$. $\square$

Corollary 10.4 (Consistency of genuinely uniform classes). Assuming $\text{Con}(\text{ZFC} + \text{I0})$, it is consistent that an ultraexacting λ has at least λ classes q such that *every* OD_{V_λ} subset of D_λ meets q in either zero or λ representatives. At the same time the bounded quotient admits a definable transversal, and the cofinal quotient admits transversals by Choice but none that is OD_{V_λ} .

Proof. Take a model of T_{sat} . Clause (ii) is exactly the first assertion, and clause (iv) gives the bounded transversal. Ambient Choice selects a representative from every member of the set Q_λ . Corollary 6.1 excludes a low-parameter ordinal definition of any such transversal. $\square$

11 Proof audit and limits of the result

11.1 The dependency graph

The essential proof has the following short chain:

$$\begin{aligned} \text{local Kunen} + \text{exacting witness} &\implies \text{no fixed small cofinal subset of } \lambda, \\ \text{no fixed small cofinal subset} &\implies \text{no fixed small family in } D_\lambda, \\ \text{ultraexactingness} &\implies \text{fixed lower-rank code and many fixed classes,} \\ \text{bounded-rank definition codes} + \text{fixed classes} &\implies \text{simultaneous full-sized traces.} \end{aligned}$$

The family theorem adds only the fixed set of fiber cardinalities. The separation theorem uses only the ordinary exacting cofinality barrier. The Icarus obstruction is a separate application to an inner model with the critical sequence already present. The equiconsistency refinement imports Input **C**, after all the new combinatorial implications have been proved in ZFC.

11.2 Checks at the main potential failure points

Ordinal parameters. We never infer that an arbitrary OD_p set is fixed merely because p is fixed. Its ordinal parameters might be moved. Instead, bounded-rank stabilization is arranged before the embedding is chosen, and the least counterexample is defined from parameters known to be fixed.

A nontransitive domain. Membership $B \in X$ does not imply $B \subseteq X$. The small-family proofs do not make that inference. They use definable union and, for a set of ordinals, the uniquely definable increasing enumeration. Pointwise image arguments are used only when an actual enumeration by fixed natural numbers belongs to X .

Singular cardinal arithmetic. For a small family of countable sets, the estimate is $\mu \cdot \aleph_0$. For a small family of thin sections, the bound $\mu \cdot \kappa$ is justified only after the embedding has forced all relevant fiber cardinals below κ . Replacing this step by “a union of $< \lambda$ small sets is small” would be invalid.

Correctness chosen in the right order. The height ζ is first made correct for all relevant bounded-rank OD_p objects, uniformly over $p \in V_{\lambda+1}$. The embedding, its fixed code b , and the classes q_t are selected

afterwards. Neither the critical sequence nor the coding parameter is used prematurely to choose the correctness height.

External versus internal cardinalities. All smallness estimates use ambient ZFC. In the Icarus argument, the inner model need not see a bijection witnessing the smallness of B . It needs only the order type of $\bigcup B$, which is absolute and lies below λ by the ambient estimate.

Equiconsistency versus implication in the same universe. The proof does not assert $\text{UEx}(\lambda) \Rightarrow \text{I0}(\lambda)$. It uses an equiconsistency theorem, followed in the appropriate direction by a preservation theorem, and explicitly proves that the new clauses hold in the resulting model.

11.3 What remains open in this continuation

The simultaneous conclusion over V_λ is settled by Theorem 5.4; it does not require $V_\lambda \subseteq \text{HOD}$. Proposition 6.6 settles the next uniform rank in the opposite direction. The remaining questions are genuinely different.

Question 11.1. Can the lower bound $|\mathcal{G}_{\text{low}}(\lambda)| \geq \lambda$ be strengthened, for example to 2^λ , under the same hypothesis? The proof produces arbitrarily large critical-point-sized families below λ ; it does not establish that its bound is optimal.

Question 11.2. Can ordinary exactingness replace ultraexactingness in the thin-section obstruction? The local obstruction already needs only exactingness, but the construction of a fixed critical-sequence class uses $j \restriction V_\lambda \in X$. An alternative way to obtain a fixed suitable class would be needed.

The supplied synthesis's small-family problem for selectors on finite subsets of Q_λ remains distinct and unresolved here, as does its cover-exacting/strongly-compact interaction problem. No conclusion about those questions is inferred from terminology shared with the present thin-section results.

11.4 Novelty and verification status

The contribution relative to the supplied documents is the replacement of finite fibers by arbitrary fibers of size $< \lambda$, the simultaneous λ -many-class conclusion over all low-rank parameters, the nonuniform small-family extension, the explicit preserved-predicate obstruction, and the stated rank-separation lemma. The low-parameter saturation theorem, which needs only ultraexactingness, is then added to the existing I0-equiconsistent coding model.

The primary-source checks support the external inputs and distinguish them from the new deductions. They do not certify worldwide priority: related consequences may be known to specialists or appear under different formulations. The proofs have been checked internally for the issues listed above, but have not been formalized in a proof assistant or independently refereed. These qualifications apply to the new results, not to a claimed proof of an unconditional inconsistency, which this report does not present.

Research conclusion

The finite-transversal boundary can be strengthened to a cardinal-sized one: at an ultraexacting λ , a definable complete section must have λ representatives in many classes, and the classes can be chosen simultaneously for all definitions with arbitrary parameters from V_λ . The parameter bound is sharp at the next rank. Even a definable small family of nondefinable thin sections is impossible.

At the same consistency strength as $I0$, this uniform saturation coexists with $V_\lambda \subseteq \text{HOD}$, while definable choice survives on the bounded quotient. This yields both a conditional inconsistency theorem and a positive relative-consistency refinement, with no claim that ultraexactingness itself is inconsistent.

A Rank and definability bookkeeping

This appendix records bounds used implicitly in the elementarity arguments. They are intentionally generous; optimizing the finite rank offset would not change any conclusion.

For an infinite limit cardinal λ , each ordinal below λ and every finite tuple of such ordinals has rank below λ . A subset of λ has rank at most λ , hence belongs to $V_{\lambda+1}$. It follows that

Object	A rank containing it
$a \subseteq \lambda$	$V_{\lambda+1}$
$D_\lambda, q \in Q_\lambda, T \subseteq D_\lambda$	$V_{\lambda+2}$
$Q_\lambda, \mathcal{F} \subseteq \mathcal{P}(D_\lambda)$	$V_{\lambda+3}$

A function whose domain is an ordinal below λ and whose values are members of any one of these displayed sets has graph of rank at most λ plus a fixed finite number. In particular, all possible bijections witnessing cardinalities $< \lambda$, all increasing enumerations of subsets of λ , and the maps used to list fiber cardinalities occur below $V_{\lambda+12}$. A height $\zeta > \lambda + \omega$ therefore contains the full collection of relevant witnesses. This proves the cardinality absoluteness used in Sections 4 and 7.

The relative hereditary class HOD_{V_λ} is the class of sets all of whose transitive-closure members, including the set itself, belong to OD_{V_λ} . It is transitive, contains V_λ , and satisfies ZF. For example, its internal power set of x is defined by separation from $\mathcal{P}(x)$ using the definable hereditary class; its internal replacement sets are obtained by ambient Replacement for the relativized formula. The resulting sets are definable from the finitely many low-rank parameters needed for x and the formula's parameters, together with the ordinal λ , and their elements remain hereditary. Finite lists of low-rank parameters can be combined into one because λ is a limit cardinal. These observations justify the closure operations used in Theorem 9.2; they do not assert global Choice in that inner model.

Finally, the canonical code order in Section 4 can be made completely explicit. At a fixed height θ , order codes first by the finite tuple length, then by the formula number, then lexicographically on the ordinal tuple. For a fixed finite length this is a wellordering. The ordinal sum of these orders over the natural numbers is still a set wellordering. Placing the height θ first gives a definable class order with a set of predecessors below every code. Thus “the least code” does not appeal to an arbitrary global choice function.

B Build and source notes

The accompanying source is self-contained apart from standard TeX Live packages. It uses the same `newpxtext/newpxmath` font family, page geometry, heading treatment, callout boxes, and forest/o-live/sage palette as the supplied reports. The colors are 30392C, 687144, 65685F, CBD0BC, and F2F4EB. The source can be compiled with two or three runs of `pdflatex`; the archive includes build instructions. No font files are included.

The bibliography distinguishes the two supplied manuscripts from primary external sources. The references to the supplied synthesis concern its finite-modification quotient and finite-valued transversal arguments; the new proofs are written in full here and do not depend on its unpublished proofs. The source versions below were checked on 18 September 2026.

References

- [1] Juan P. Aguilera, Joan Bagaria, and Philipp Lücke. *Large cardinals, structural reflection, and the HOD Conjecture*. arXiv:2411.11568. Version consulted: <https://arxiv.org/html/2411.11568v4>. Especially Lemmas 2.3 and 3.2 and Theorem 2.10.
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- [3] Kenneth Kunen. Elementary embeddings and infinitary combinatorics. *Journal of Symbolic Logic* 36(3) (1971), 407–413. doi:10.2307/2269948.
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- [5] *Large cardinals at the inconsistency frontier: a synthesis*. User-supplied research manuscript, 18 September 2026. File: `Large_Cardinals_Synthesis.tex` in `Cardinals2.zip`. Especially the finite-choice and finite-transversal development in Sections 8–9, and the subsequent consistency discussion.
- [6] *Large cardinals at the inconsistency frontier: a unified assessment*. User-supplied research manuscript, 18 September 2026. File: `Large_Cardinals_Unified_Report.tex` in `Cardinals2.zip`.